

Take-Home Assessment

Start Here

Everything you need lives in this repository: github.com/AmmarKeon/cluster_assessment

Thank you for taking the time. This document is the map; the real contracts are the two PDFs inside the repository, one per task. Read them fully before you write code.

First thing: fork the repository

Fork github.com/AmmarKeon/cluster_assessment to your own GitHub account and do all of your work there. Do not clone the original and work off it, do not open pull requests against it. Your fork is your workspace — commit as you go, exactly like you would on the job. When you finish, send us your fork link to the contact at the bottom.

What is inside

Folder	Contents
task1/	the Task-1 brief (TASK1.pdf) + its own dataset/ of CSVs
task2/	the Task-2 brief (TASK2.pdf) + its own dataset/ of CSVs

The two tasks are fully independent: separate data, separate deliverables, graded separately. The datasets look similar on purpose — do not assume one task's files tell you anything about the other's scope. Each CSV set carries a `_manifest.json` with row counts and checksums; verify your copy before starting.

Task 1 – the short version

Three systems recorded the same pharmacy orders and none of them agree with each other. Unify them: map every supplier-side account name to the real pharmacy behind it, recover the pharmacy locations that our records lost (the evidence is in the invoices, if you look where nobody usually looks), build one clean revenue ledger, and answer four ranking questions – areas, pharmacies, suppliers. Then wrap it in a small local website – one double-click on a `run.bat` – so the results can actually be browsed and filtered, not just read. **Full specification:** `task1/TASK1.pdf`.

Task 2 – the short version

A field rep has 16 pharmacies to visit tomorrow, in one area. Decide the order they should be visited in. The history tells you how long a stop takes, when each pharmacy is best visited, and how long drives really feel – nothing is handed to you. Produce the itinerary with honest timings, prove it fits inside the day, handle whatever does not work out, and ship a small page that draws the route, launched with a run.bat. **Full specification:** `task2/TASK2.pdf`.

How we grade, in one breath

We care about verifiable work over fluent stories: every number reproducible by one command, every claim backed by a check, every unknown measured instead of invented. Offline code only – no network calls, no model APIs inside your submission. Using AI assistants is expected; keep an honest `AI_DISCLOSURE.md` of what you used and what you rejected. Both PDFs contain the full rules, deliverable shapes, and grading weights.

Questions

Anything about this assessment: Ammar Yasser – +20 127 547 5215 (WhatsApp). Ask early, ask specific; we answer fast.